Recursion Practice Assignment

Total Questions: 26 | Language: Java | Topic: Recursion

Questions 1–6: Basic Recursion

Q1. Print Numbers from 1 to N

Problem Statement: Given an integer (N) print all integers from 1 to (N)in increasing order using recursion.

Input Format: A single integer (N)

Output Format: Print the numbers from 1 to (N) separated by spaces.

Example

```
Input:
5

Output:
1 2 3 4 5
```

Q2. Print Numbers from N to 1

Problem Statement: Given an integer (N), print all integers from (N)to 1 in decreasing order using recursion.

Input Format: A single integer (N)

Output Format: Print the numbers from (N)to 1, separated by spaces.

Example

```
Input:
5

Output:
5 4 3 2 1
```

Q3. Print Even Numbers from 1 to N

Problem Statement: Given an integer (N) print all even numbers between 1 and (N)in increasing order using recursion.

Input Format: A single integer (N)

Output Format: Print all even numbers from 1 to (N) separated by spaces.

Example

```
Input:
10
```

```
Output:
2 4 6 8 10
```

Q4. Print Odd Numbers from 1 to N

Problem Statement: Given an integer (N) print all odd numbers between 1 and (N)in increasing order using recursion.

Input Format: A single integer (N)

Output Format: Print all odd numbers from 1 to (N) separated by spaces.

Example

```
Input:
10

Output:
1 3 5 7 9
```

Q5. Sum of First N Natural Numbers

Problem Statement: Given a positive integer (N) calculate the sum of the first (N)natural numbers using recursion.

Input Format: A single positive integer (N)

Output Format: Print the sum of the first (N)natural numbers.

Example

```
Input:
5

Output:
15
```

Explanation: (1+2+3+4+5=15).

Q6. Factorial of a Number

Problem Statement: Given a non-negative integer (N) calculate its factorial using recursion. The factorial of (N)is the product of all positive integers from 1 to (N) By definition, (0! = 1).

Input Format: A single non-negative integer (N)

Output Format: Print the factorial of (N)

Example

```
Input:
5

Output:
120
```

Explanation: (5! = 5*4*3*2*1=120)

Questions 7–26: Recursion Practice

Q7. Reverse a String

Problem Statement: Given a string, reverse it using recursion without using a built-in method that directly reverses the string.

Input Format: A single string (S).

Output Format: Print the reversed string.

Example

```
Input:
hello

Output:
olleh
```

Q8. Check Whether a String Is a Palindrome

Problem Statement: Given a string (S)determine whether it reads the same forward and backward. Use recursion to check the string. Treat uppercase and lowercase letters as different characters.

Input Format: A single string (S).

Output Format: Print `Palindrome` if the string is a palindrome; otherwise, print `Not Palindrome`.

Example 1

```
Input:
madam

Output:
Palindrome
```

Example 2

```
Input:
hello

Output:
Not Palindrome
```

Q9. Reverse a Number

Problem Statement: Given a non-negative integer (N) reverse its digits using recursion.

Input Format: A single non-negative integer (N)

Output Format: Print the reversed number. Leading zeros in the reversed result should not be displayed.

Example

```
Input:
1234

Output:
4321
```

Q10. Sum of Digits

Problem Statement: Given a non-negative integer (N) calculate the sum of its digits using recursion.

Input Format: A single non-negative integer (N)

Output Format: Print the sum of the digits.

Example

```
Input:
1234

Output:
10
```

Explanation: (1+2+3+4=10).

Q11. Product of Digits

Problem Statement: Given a non-negative integer (N) calculate the product of its digits using recursion. For this problem, the product of the digits of 0 is 0.

Input Format: A single non-negative integer (N)

Output Format: Print the product of the digits.

Example

```
Input:
234

Output:
24
```

Explanation: (2times3times4=24).

Q12. Count the Digits of a Number

Problem Statement: Given a non-negative integer (N) count the number of digits in it using recursion. Treat 0 as a one-digit number.

Input Format: A single non-negative integer (N)

Output Format: Print the number of digits.

Example

```
Input:
12345

Output:
5
```

Q13. Power of a Number

Problem Statement: Given two non-negative integers (a) and (b), calculate (a^b) using recursion. Assume the result fits within the Java `long` data type.

Input Format: Two space-separated integers (a) and (b).

Output Format: Print (a^b).

Example

```
Input:
2 5

Output:
32
```

Explanation: (2^5=2times2times2times2times2=32).

Q14. Fibonacci Series

Problem Statement: Given a positive integer (N) print the first (N)terms of the Fibonacci series using recursion. The series starts with 0 and 1, and each subsequent term is the sum of the previous two terms.

Input Format: A single positive integer (N)

Output Format: Print the first (N)Fibonacci terms, separated by spaces.

Example

```
Input:
7

Output:
0 1 1 2 3 5 8
```

Q15. GCD of Two Numbers

Problem Statement: Given two non-negative integers (A) and (B), find their greatest common divisor (GCD) using recursion and the Euclidean algorithm. At least one number will be positive.

Input Format: Two space-separated non-negative integers (A) and (B).

Output Format: Print the GCD of the two numbers.

Example

```
Input:
12 18

Output:
6
```

Q16. LCM of Two Numbers

Problem Statement: Given two positive integers (A) and (B), find their least common multiple (LCM). Use a recursive GCD function to help calculate the answer.

Input Format: Two space-separated positive integers (A) and (B).

Output Format: Print the LCM of the two numbers.

Example

```
Input:
4 6

Output:
12
```

Explanation: 12 is the smallest positive integer divisible by both 4 and 6.

Q17. Decimal to Binary

Problem Statement: Given a non-negative decimal integer (N) convert it into its binary representation using recursion. Do not use Java's built-in decimal-to-binary conversion method.

Input Format: A single non-negative integer (N)

Output Format: Print the binary representation of (N) For (N=0), print `0`.

Example

```
Input:
10

Output:
1010
```

Q18. Binary to Decimal

Problem Statement: Given a string containing only binary digits (`0` and `1`), convert it into its decimal equivalent using recursion.

Input Format: A single binary string (S).

Output Format: Print the decimal equivalent of the binary number.

Example

```
Input:
1011

Output:
11
```

Explanation: (1011_2=8+0+2+1=11_{10}).

Q19. Count Occurrences of a Character in a String

Problem Statement: Given a string (S) and a character (C), count how many times (C) appears in (S) using recursion. Treat uppercase and lowercase characters as different.

Input Format:

- First line: A string (S).
- Second line: A character (C).

Output Format: Print the number of occurrences of (C) in (S).

Example

```
Input:
programming
g

Output:
2
```

Q20. Remove a Character from a String

Problem Statement: Given a string (S) and a character (C), remove all occurrences of (C) from the string using recursion. Treat uppercase and lowercase characters as different.

Input Format:

- First line: A string (S).
- Second line: A character (C).

Output Format: Print the resulting string after removing all occurrences of (C).

Example

```
Input:
banana
a

Output:
bnn
```

Q21. Replace a Character in a String

Problem Statement: Given a string (S) and two characters (C_1) and (C_2), replace every occurrence of (C_1) with (C_2) using recursion. Treat uppercase and lowercase characters as different.

Input Format:

- First line: A string (S).
- Second line: Two space-separated characters (C_1) and (C_2).

Output Format: Print the modified string.

Example

```
Input:
banana
a o

Output:
bonono
```

Q22. Check Whether an Array Is Sorted

Problem Statement: Given an integer array, determine whether it is sorted in non-decreasing order using recursion. An array is sorted if every element is less than or equal to the element following it.

Input Format:

- First line: An integer (N) the size of the array.
- Second line: (N)space-separated integers.

Output Format: Print `Sorted` if the array is sorted; otherwise, print `Not Sorted`.

Example 1

```
Input:
5
1 2 2 4 7

Output:
Sorted
```

Example 2

```
Input:
4
1 5 3 7

Output:
Not Sorted
```

Q23. Find the Maximum Element in an Array

Problem Statement: Given an integer array of size (N) find its maximum element using recursion.

Input Format:

- First line: An integer (N) the size of the array.
- Second line: (N)space-separated integers.

Output Format: Print the maximum element in the array.

Example

```
Input:
5
3 9 2 15 6

Output:
15
```

Q24. Find the Minimum Element in an Array

Problem Statement: Given an integer array of size (N) find its minimum element using recursion.

Input Format:

- First line: An integer (N) the size of the array.
- Second line: (N)space-separated integers.

Output Format: Print the minimum element in the array.

Example

```
Input:
5
3 9 2 15 6

Output:
2
```

Q25. Linear Search Using Recursion

Problem Statement: Given an integer array and a target value (X), search for (X) in the array using recursive linear search. Return the index of its first occurrence using zero-based indexing. If the target is not present, return `-1`.

Input Format:

- First line: Two space-separated integers (N)and (X), where (N)is the array size and (X) is the target.
- Second line: (N)space-separated integers.

Output Format: Print the zero-based index of the first occurrence of (X), or `-1` if it is absent.

Example 1

```
Input:
5 7
3 7 2 7 9

Output:
1
```

Example 2

```
Input:
4 6
1 2 3 4

Output:
-1
```

Q26. Binary Search Using Recursion

Problem Statement: Given a sorted integer array in non-decreasing order and a target value (X), find the index of (X) using recursive binary search. Return the index of the target's first occurrence if it appears more than once. Use zero-based indexing. If the target is absent, return `-1`.

Input Format:

- First line: Two space-separated integers (N)and (X), where (N)is the array size and (X) is the target.
- Second line: (N)space-separated integers in sorted order.

Output Format: Print the zero-based index of the first occurrence of (X), or `-1` if it is absent.

Example 1

```
Input:
6 8
1 3 5 8 8 10

Output:
3
```

Example 2

```
Input:
5 4
1 2 3 5 6

Output:
-1
```